UNIVERSITY OF SCIENCE - VNUHCM

FACULTY OF INFORMATION TECHNOLOGY

SOFTWARE TESTING

HW6 - PERFORMANCE TESTING

Software Testing Project Report

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1 Group Information

Group ID: 07

Member Name	Student ID	Assigned Features	Status
Cao Uyển Nhi	22127310	Homepage - Category - Product	Done
Lưu Thanh Thuý	22127410	Register	Done
Nguyễn Phước Minh Trí	22127424	Homepage - Sort - Search	Done
Võ Lê Việt Tú	22127435	Sign In - Invoice	Done
Trần Thị Cát Tường	22127444	Homepage - Contact	Done

2 Server / PC Configuration

Component	**Specification**
OS	macOS Sonoma 14.5
CPU	Apple M3, 10 cores
RAM	16 GB
Disk	512 GB SSD
JMeter Version	Apache JMeter 5.6.3
Java Version	OpenJDK 11
Browser	Chrome (for script recording)
Website Under Test	Localhost (Registration feature)

3 Test Scenario

Description: Test the **Registration** function of the application under various load conditions using CSV data for multiple user accounts.

User Journey:

- **GET /** – Load Homepage
- **GET /auth/login** – Load Login Page
- **POST /register** – Submit Registration Form (data from CSV)

4 Data-Driven Setup

- **CSV Data File:** `registration_info.csv`
- **Fields:** first_name,last_name,address,city,country,dob,email,password,phone,postcode,state
- **JMeter Element:** CSV Data Set Config
- **Configuration:**
  - **Recycle on EOF:** True
  - **Stop Thread on EOF:** False
  - **Sharing Mode:** All Threads

5 Performance Testing Techniques

5.1 Load Testing

Objective: Verify system can handle normal expected load without performance degradation.

Configuration:

- **Thread Group:** 50 users
- **Ramp-Up Time:** 250s
- **Loop Count:** 1
- **Assertions:** HTTP 200, response time < 2000ms

Expected Result:

- p95 response time < 2s
- Error rate < 1%
- Throughput stable

Actual Result:

- p95 response time: 1.8s
- Error rate: 0%
- Throughput: Stable

Youtube video link: Load Testing

5.2 Stress Testing

Objective: Identify system breaking point by gradually increasing users until performance degrades.

Configuration (Ultimate Thread Group):

Start Threads	Initial Delay	Startup	Hold	Shutdown
50	0	30	240	10
150	100	30	150	10
200	200	30	90	10
200	300	30	60	10

Expected Result:

- System should maintain < 5% error rate until near breaking point.
- Gradual response time increase is acceptable.

Actual Result:

- Breakpoint reached at ~600 concurrent users.
- Error rate spiked to 7% at 600 users.
- p95 response time exceeded 5s at max load.

Youtube video link: Stress Testing

5.3 Spike Testing

Objective: Test system behavior with sudden, extreme load increase and recovery.

Configuration (Ultimate Thread Group):

Start Threads	Initial Delay	Startup	Hold	Shutdown
50	0	30	420	10
550	90	5	60	5
550	210	5	60	5
550	330	5	60	5

Expected Result:

- System should recover quickly after spike.
- No persistent errors after returning to baseline.

Actual Result:

- System recovered in ~15s after each spike.
- Temporary error rate spike to 5% during spike events.

Youtube video link: Spike Testing

6 Report Viewers Used

1. **View Results Tree** – For detailed request/response verification.
2. **Summary Report** – Aggregated metrics table.
3. **Response Time Graph** – Visualizing p95 trends over time.

7 Step-by-Step Instructions

- **Record Test Script** using HTTP(S) Test Script Recorder while performing registration.
- **Parameterize Data** by adding CSV Data Set Config and linking variables in POST request.
- **Add Assertions** for response code and max response time.
- **Configure Thread Group** according to each test type (Load, Stress, Spike).
- **Add Report Viewers** (View Results Tree, Summary Report, Response Time Graph).
- **Run Test** and monitor metrics.
- **Save Results** in `.jtl` files and export graphs for reporting.
- **Analyze** and document findings.

8 AI / LLM Application and Prompts

How AI was applied:

- **Test Plan Design:** Used ChatGPT to suggest optimal thread group configurations for Load, Stress, and Spike Testing scenarios.
- **Parameterization Logic:** AI provided guidance for setting up CSV Data Set Config for dynamic test data.
- **Bottleneck Analysis:** After running the tests, AI helped interpret p95 response time patterns and identify potential database bottlenecks.
- **Prompt Engineering:** AI generated reusable prompts for creating .jmx thread group setups and debugging common JMeter errors.

Benefits:

- Reduced setup time for test configurations.
- More accurate and consistent testing steps.
- Automated documentation generation for reporting.

Example Prompts Used:

1. *"Suggest optimal Ultimate Thread Group configuration for stress testing starting at 50 users up to 1000 users with 90s hold per step."*
2. *"Generate a JMeter CSV Data Set Config setup for registration with 100 unique accounts."*
3. *"Analyze JMeter Summary Report results to identify bottlenecks in response time."*
4. *"Explain why my Ultimate Thread Group chart drops to zero between steps and how to fix it."*
5. *"Write a detailed performance testing report in markdown for Load, Stress, and Spike testing using my given configuration and results."*

9 Self-Assessment

Criteria	Description	Max Points	Self Assessment
Load testing	Report: 1.0 TestCases, BugReport: 0.5 Script, Data: 0.5 Video: 1.0	**3.0**	**3.0**
Stress testing	Report: 1. TestCases, BugReport: 0.5 Script, Data: 0.5 Video: 1.0	**3.0**	**3.0**
Spike testing	Report: 1. TestCases, BugReport: 0.5 Script, Data: 0.5 Video: 1.0	**3.0**	**3.0**
Use of AI Tools	Prompt transparency, critical validation, added value	**1.0**	**1.0**
Total		**10.0**	**10.0**